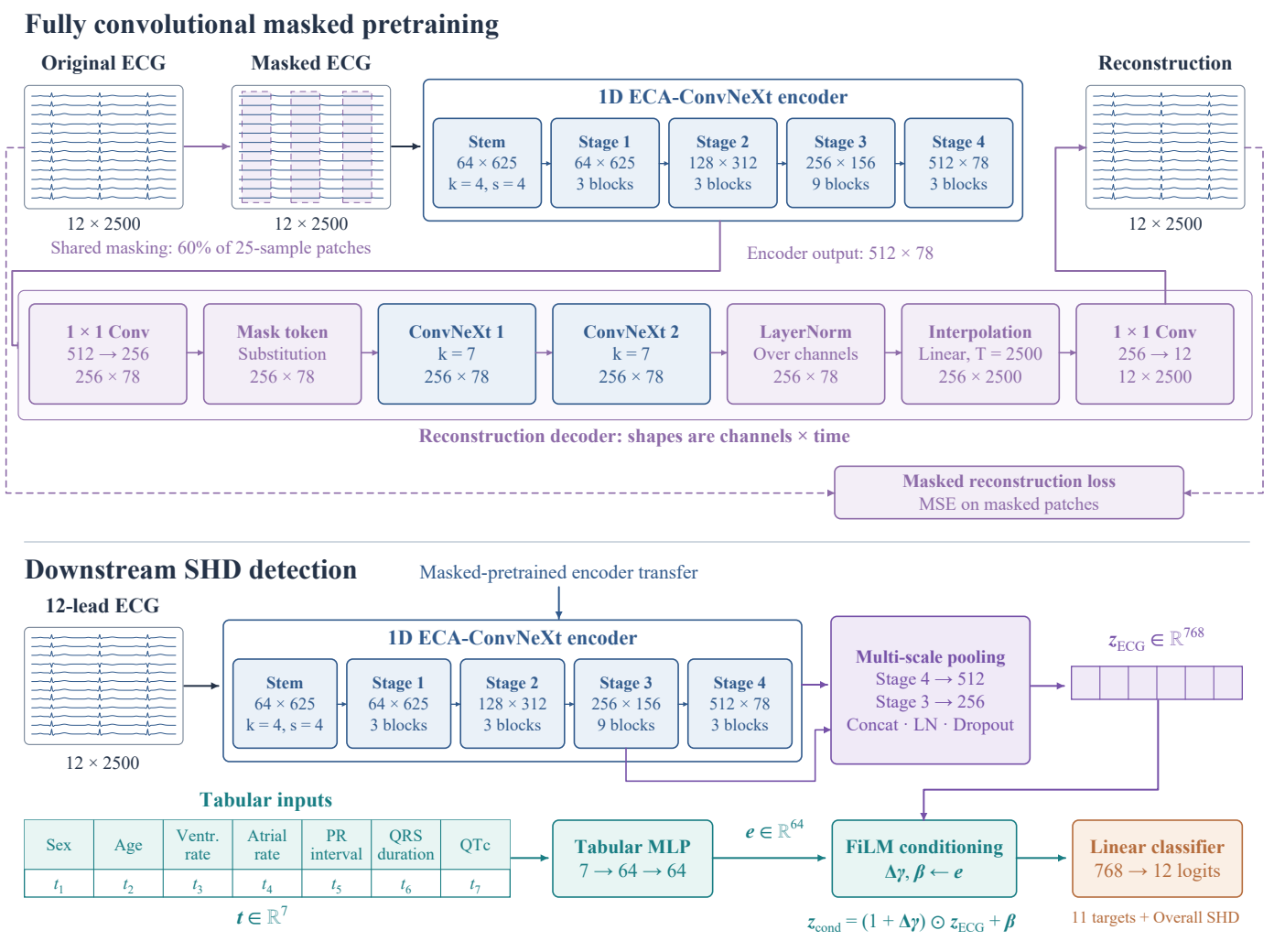

Fully convolutional masked pretraining
Original ECG
Masked ECG
Reconstruction
1D ECA-ConvNeXt encoder
Stem
64 × 625
k = 4, s = 4
Stage 1
64 × 625
3 blocks
Stage 2
128 × 312
3 blocks
Stage 3
256 × 156
9 blocks
Stage 4
512 × 78
3 blocks
12 × 2500
12 × 2500
12 × 2500
Shared masking: 60% of 25-sample patches
Encoder output: 512 × 78
1 × 1 Conv
512 → 256
256 × 78
Mask token
Substitution
256 × 78
ConvNeXt 1
k = 7
256 × 78
ConvNeXt 2
k = 7
256 × 78
LayerNorm
Over channels
256 × 78
Interpolation
Linear, T = 2500
256 × 2500
1 × 1 Conv
256 → 12
12 × 2500
Reconstruction decoder: shapes are channels × time
Masked reconstruction loss
MSE on masked patches
Downstream SHD detection
Masked-pretrained encoder transfer
12-lead ECG
12 × 2500
1D ECA-ConvNeXt encoder
Stem
64 × 625
k = 4, s = 4
Stage 1
64 × 625
3 blocks
Stage 2
128 × 312
3 blocks
Stage 3
256 × 156
9 blocks
Stage 4
512 × 78
3 blocks
Multi-scale pooling
Stage 4 → 512
Stage 3 → 256
Concat · LN · Dropout
$z_{ECG} \in \mathbb{R}^{768}$
Tabular inputs
Sex
Age
Ventr. rate
Atrial rate
PR interval
QRS duration
QTc
t_1
t_2
t_3
t_4
t_5
t_6
t_7
$t \in \mathbb{R}^7$
Tabular MLP
7 → 64 → 64
$e \in \mathbb{R}^{64}$
FiLM conditioning
$\Delta\gamma, \beta \leftarrow e$
$z_{cond} = (1 + \Delta\gamma) \odot z_{ECG} + \beta$
Linear classifier
768 → 12 logits
11 targets + Overall SHD